

System Design & Backend Architecture

Chapter One — MVP

Tech Stack

Frontend

- Next.js
- React Native (Expo)
- TypeScript

Backend

- NestJS
- Prisma ORM

Database

- PostgreSQL

Realtime

- Socket.IO
- Redis

Storage

- Cloudflare R2

AI

- OpenAI (Moderation)

Deployment

- Docker
- Coolify

Monitoring

- Sentry
- PostHog
- Better Stack

Backend Modules

```
Auth
Users
Profiles
Matching
Introductions
Chat
Moderation
Notifications
Storage
Admin
```

Database Tables

```
users
profiles
photos
voice_intros

matches
introductions

messages

interests
user_interests

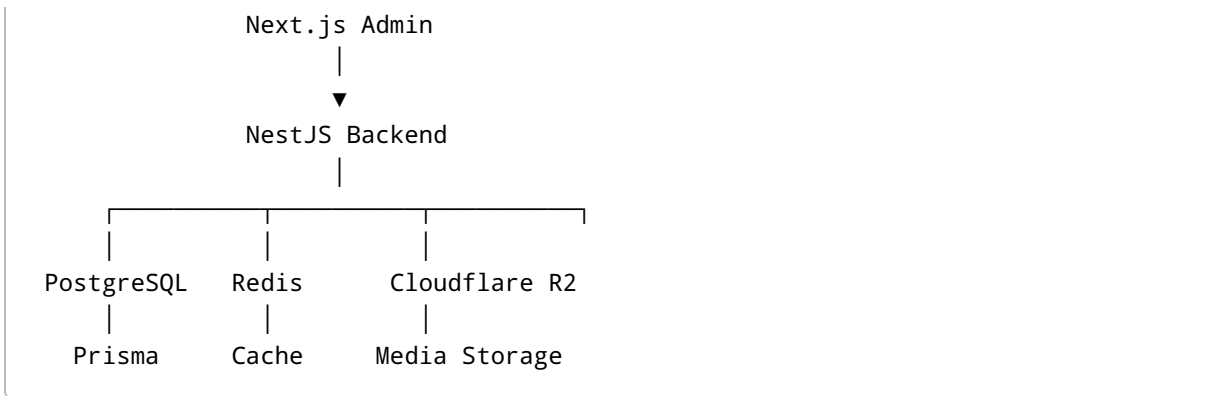
reports
blocks

notifications

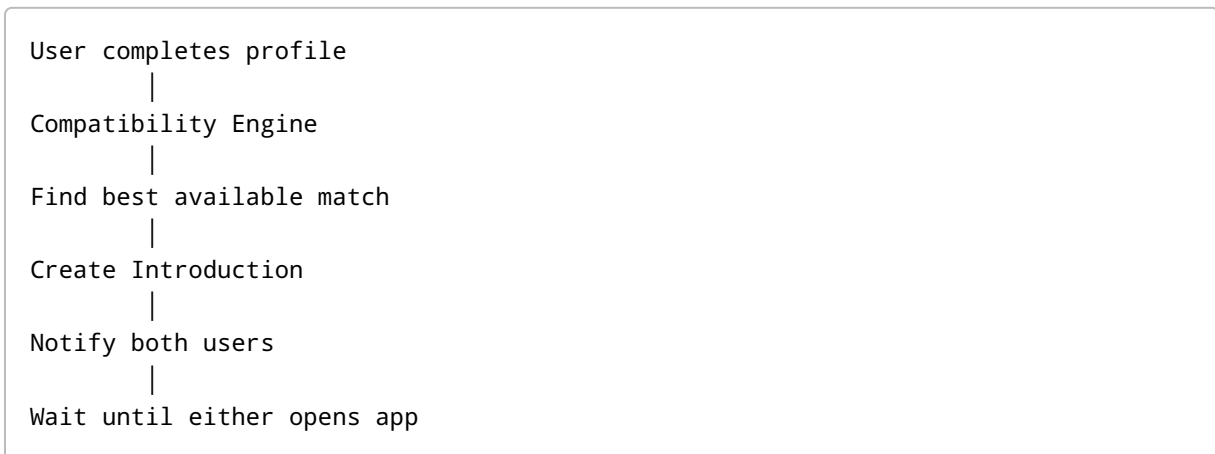
device_tokens
```

High-Level Architecture

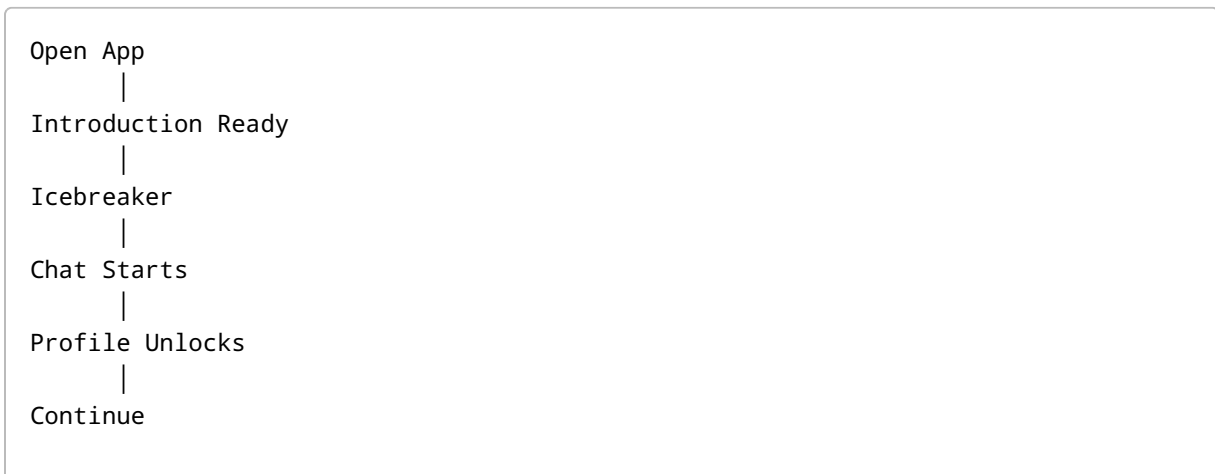
```
React Native
|
```



Matching Flow



Introduction Flow



or
End Introduction

Chat Flow

```
graph TD; RN[React Native] --> SI[Socket.IO]; SI --> NG[NestJS Gateway]; NG --> PG[PostgreSQL];
```

Redis handles:

- Online presence
 - Typing status
 - Socket sessions
-

Media Flow

```
graph TD; C[Client] --> SU[Signed Upload URL]; SU --> CR[Cloudflare R2]; CR --> P[Store URL in PostgreSQL];
```

AI Moderation

Moderate:

- Messages
- Images
- Voice transcripts (optional)

Actions:

- Warning
 - Hide content
 - Temporary suspension
 - Permanent ban
-

Security

- JWT Authentication
 - Refresh Tokens
 - Encrypted passwords
 - HTTPS only
 - Rate limiting
 - Signed upload URLs
 - Government ID verification
-

Future Expansion

- Recommendation Engine
- Video Calls
- AI Matchmaking
- Couple Features
- Premium Subscription
- Analytics Dashboard
- Admin Moderation Console
- Event-driven Microservices (only if scale demands it)